

Tushar Alase

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EDUCATION

Purdue University – West Lafayette, IN

Aug 2026 – May 2028 (Expected)

Master of Science in Electrical and Computer Engineering

Savitribai Phule Pune University – Pune, India

Oct 2020 – Apr 2023

Bachelor of Engineering in Electronics and Telecommunication Engineering Honours: Internet of Things – GPA: 9.11/10.00

EXPERIENCE

IoT & AI Product Consultant / Freelancer

Mar 2026 – Jul 2026

AI Unika

Remote

- Advised cross-functional teams on platform selection, system architecture, deployment planning, and product integration across 4 edge-AI and IoT products.
- Planned the integration of Raspberry Pi and NVIDIA Jetson platforms with cameras, sensors, LiDAR, and telemetry systems based on product and deployment requirements.

Senior Embedded System Developer

Mar 2024 – Feb 2026

Cyamsys Technologies

Noida, India

- Architected an end-to-end FOTA and device-management system supporting 150 devices and achieving 95% first-attempt update success, with retry and rollback mechanisms recovering the remaining devices.
- Developed production Embedded C/C++ firmware across 5 industrial product categories, implementing peripheral interfaces, communication protocols, automation logic, and real-time control functions.
- Automated firmware testing using Python tools, reducing validation time by 50%—from 3 hours to 90 minutes—across 5 protocols and 6 projects.

Embedded Developer Intern → Product Manager / Technical Lead

Oct 2022 – Aug 2023

FIKKAR Innovations

Pune, India

- Advanced from embedded intern to product leadership while guiding 5 prototypes from concept through testing and demonstration.
- Coordinated a 6-member team across 4 functional groups, including PCB design, embedded firmware, mobile applications, and enclosure design and procurement.

SELECTED PROJECTS

IIoT Data Logger & Remote Device Management System

- Designed a connected data-logger platform with BLE-based monitoring, device configuration, multi-protocol telemetry, and remote diagnostics.
- Architected Wi-Fi/GSM FOTA workflows with retry and rollback support for reliable remote firmware maintenance.

Advanced Driver Assistance System for Road Anomalies

- Developed a YOLOv8-based computer-vision pipeline for real-time road-anomaly detection and driver alerts.
- Integrated predictive road mapping and V2V/V2X communication to share detected hazards with nearby connected vehicles.

AutOmelette Articulated Cooking Robot

- Developed embedded control workflows for a 6-DOF robotic arm using motor actuation, sensor feedback, and automated cooking sequences.
- Integrated location-aware recipe adaptation and IoT functionality; earned recognition in 3 engineering competitions.

TECHNICAL SKILLS

Programming: C, C++, Embedded C, Python

Embedded & Edge Platforms: ESP32/ESP-IDF, STM32, MSP430/MSPM0, ATmega4809, ARM Cortex-M, Raspberry Pi, NVIDIA Jetson, RTOS, Embedded Linux

Communication & IoT: UART, SPI, I2C, RS-485, Modbus, BLE, Wi-Fi, LoRa, GSM/4G, MQTT, HTTP/FTP, telemetry, remote diagnostics

Tools & Hardware: Git/GitHub, Code Composer Studio, STM32CubeIDE, PlatformIO, KiCad, JTAG, logic analyzers, sensor integration, board bring-up

SELECTED AWARDS

Technology Infusion Grand Challenge Asia Funding, La Trobe University, 2022 | **Audience Choice Award**, Circuit Digest, 2021 | **Winner**, Abhiyantrix, 2023